SAAS INTEGRATION BLUEPRINT

Instagram Insights API Architecture

Automated, zero-lag data fetching protocol to connect creators and brands natively, eliminating the need for manual screenshot submissions.

THE CONSTRAINT

Relying exclusively on creators to manually paste profile URLs restricts our system to public data scraping. This methodology not only violates Meta's strict privacy policies but also completely fails to fetch crucial deep insights such as Reach, Impressions, and detailed Audience Demographics required by brands.

THE INNOVATION

Integrating the **Official Meta Graph API**. By replacing manual inputs with a seamless OAuth button, creators securely grant platform access. This constructs a verified, real-time analytics pipeline directly into the brand dashboard, ensuring 100% data integrity and compliance.

DATA AUTHENTICATION PIPELINE

Implementation Mechanics

1 Meta Developer Setup (Free Tier)

APPROVED

- **App Initialization:** Establish a verified 'Business' application within the developers.facebook.com portal to access Graph API endpoints natively.
- **Mandatory Scopes:** The system requires `instagram_basic`, `instagram_manage_insights`, and `pages_show_list` to fetch deep analytics efficiently.
- **App Review Protocol:** Progression to public launch mandates a visual screencast submitted to Meta, demonstrating exactly how SyncHub utilizes and displays this creator data on the dashboard.

2 Authentication & Token Logic

By eliminating URL pasting, the creator is prompted with a localized login flow. A critical prerequisite is that the creator possesses an **Instagram Professional or Creator Account** intrinsically linked to a Facebook Page.

Short-Lived Token

Captured instantaneously upon successful OAuth login. This temporary key possesses a highly restricted validity period of strictly 1 to 2 hours.

Long-Lived Token (Core)

The backend architecture programmatically swaps the initial token for an extended variant. This grants a crucial **60-Day** window for seamless background data syncing.

ENDPOINT ARCHITECTURE

```
GET https://graph.facebook.com/v19.0/{ig-user-id}/insights?
metric=impressions,reach,profile_views&
period=day&access_token={long-lived-token}
```

Zero-Lag Caching Strategy

Meta imposes a stringent rate limitation policy (approximately 200 requests per hour, per user). To guarantee that the brand dashboard renders instantaneously with **absolute zero lags**, the architecture strictly forbids fetching live API data during a page load. Instead, we implement an advanced **Cron Job Caching** infrastructure.

Followers Base Count Stored in MySQL Primary Database	EVERY 12 HRS
Reach & Impressions Cached via Redis / Stats Cluster	EVERY 24 HRS
Latest 5 Posts Analytics Upserted into Media Metrics Table	EVERY 6 HRS

Automated Token Renewal Protocol

Because long-lived tokens naturally expire after 60 days, a specialized backend daemon must be deployed. This script will automatically hit the `refresh_access_token` endpoint precisely 10 days prior to expiration. This creates a self-sustaining loop, ensuring uninterrupted analytics flow for brands without ever requiring the creator to manually re-authenticate.